MIL-A-19401(Aer)
21 March 1956

MILITARY SPECIFICATION

ACTUATOR , ELECTROMECHANICAL, ROTARY, FOR AS-473/ASB-1 (BuOrd Drawing 790459, 1332290, Azimuth Torque Amplifier)

1. SCOPE

1.1 This specification covers the equipment, material, and performance requirements to be used in the manufacture of a Rotary Electromechanical Actuator for use in a radar scanning antenna. One type and one grade only is specified.

2. APPLICABLE SPECIFICATIONS, STANDARDS, DRAWINGS, AND PUBLICATIONS

2.1 The following specifications, standards, drawings, and publications of the issue in effect on date of invitation for bids form a part of this specification.

2.1.1 Specifications

Military

MIL-P-116	Preservation, Methods of

2.1.2 Standards

Military

MIL-STD-129	Military Standard Marking of Shipments

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2.1.3 Drawings

Bureau of Ordnance, Department of the Navy

List of Drawings, LD 272239, Azimuth Torque Amplifier and all drawings and specifications: listed thereon.

2.1.4 Publications

Bureau of Ordnance, Department of' the Navy

NAVORD OP 400	Manufacture and Inspection of Ordnance Material, General Specification for
NAVORD OD 8176	Paragraphs applicable to this equipment

(Copies Of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the Procuring Agency or directed by the Contracting Officer

2.1.5 Other Publications

National Bureau of Standards Publications

Handbook H28	Screw-Thread Standards for Federal Services

(Copies of H28 may be obtained upon application accompanied by money order, cupon, or cash to the Superintendent of Documents, Government Printing Office, Washington 25, D.C.)

3. REQUIREMENTS

3.1 Type Tests - Type tests as specified in Section 4 of this specification shall be made on representative sample of production assemblies.

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3.2 General - Equipment furnished under this specification shall meet applicable requirements listed in NAVORD OD 8176. In cases of conflict, this specification shall apply.

3.3 Material - Materials shall conform to applicable specifications as specified herein and on the drawings. When materials are used which are not specifically designated they shall be of the best commercial quality and entirely suitable for the purpose.

3.4 Design and Construction - The assembly shall conform to and be approved in accordance with this specification and applicable specifications standards, drawings, and publications.

3.5 Interchangability - Interchangeability shall be insured by conformance with dimensional limitations set forth on cutline drawing, BuOrd number 776762.

3.6 Nameplates - A nameplate, permanently and legibly marked and made in accordance with Buord drawing 776416 shall be attached to each assembly.

3.7 Performance

3.7.1 Torque - The torque required to slip the clutch as measured at the output gear, BuOrd drawing 837495, shall be as specified below; and the drive motor, Buord drawing 776605, shall not stall.

3.7.1.1 No torque shall be produced at the output gear when the C-shaped lever, BuOrd drawing 760038, is in its neutral position. (The neutral position is that position at which the C-shaped lever is engaged with the roller, BuOrd drawing 837526, and no torque is produced at the output gear.)

3.7.1.2 A torque of 4 to 7 inch-pounds shall be produced when the C-shaped lever is moved 0.046 ± .005 inch (measured-at roller, BuOrd drawing 837526) in both directions from its neutral position.

3.7.1.3 No torque shall be produced at the output gear when the straight lever, BuOrd drawing 760040, is in its neutral position. (Neutral position is that position at which the straight lever is engaged with the roller, BuOrd drawing 837526 and no torque is produced at the output gear, BuOrd drawing 8837495.)

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3.7.1.4 A torque of 10 ± 2 inch-pounds shall be produced when the straight lever is moved 0.125 ± .005 inch (measured at roller, BuOrd drawing 837526) in both directions from its neutral position.

3.7.2 Switches

3.7.2.1 Scan Direction Selector Switch (S505) - When the C-shaped lever is engaged with the roller, BuOrd throwing 837526, the normally open contact of switch S505 (see BuOrd drawings 790458 and 790459) shall be closed and the normally closed contact shall be open.

3.7.2.1.1 When the C-shaped lever is not engaged with the roller, the normally open contact of switch S505 shall be open and the normally closed contact shall be closed.

3.7.2.2 Scan Direction Switch (S506) - When the straight lever, BuOrd drawing 760040, is engaged with the roller and is moved from its neutral position in a direction towards the nameplate, the normally closed contact of switch S506 shall be open and the normally open contact shall be closed.

3.7.2.2.1 When the straight lever is moved from its neutral position in a direction away from the nameplate, the normally closed contact of switch S506 shall be closed and the normally open contact shall be open.

3.8 Environmental Conditions

3.8.1 Temperature - The equipment shall conform with all requirements of this specifications under temperature conditions as specified in NAVORD OD 8176 except that the temperature range shall be modified to read -55°C to +71°C (-67°F to +160°F).

3.8.1.1 To determine compliance with 3.8.1, each type test sample shall be subjected to one (1) cycle of inspection tests at maximum and minimum temperatures.

3.8.2 Humidity - The equipment shall conform with all requirements of this specification under humidity conditions as specified in OD 8176.

3.8.2.1 To determine compliance with 3.8.2, each type test sample shall be subjected to one (1) cycle of inspection tests at any combination of temperature and humidity that will alternately form water and frost on the unit.

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3.8.3 Vibration - The equipment shall be subjected to vibration conditions as specified in OD 8176 except that the total excursion shall not exceed 0.031 inch.

3.8.4 Shock - The equipment shall not suffer damage or subsequently to provide the performance specified herein when it is mounted as normally used and is subjected to six (6) impact shocks of log's, each shock impulse having a time duration of 11 ± 1 milliseconds. See BuOrd drawing 776762 for normal mounting. Two (2) shocks shall be applied in each of the three (3) principal axes, one (1) shock in each direction.

3.9 Lubrication - Lubrication shall be in accordance with insturctions on BuOrd drawing 790459.

3.10 Workmanship - Workmanship shall be in accordance with the best instrument practice. Particular attention shall be given to machined surfaces to insure conformance with drawing limitations and against superficial defects. Attention shall also be given to painted areas to insure clean and scratch-free surface.

4. SAMPLING, INSPECTION, AND TEST PROCEDURE

4.1 General Instructions

4.1.1 Responsibility for Tests - The contractor shall furnish all samples, if required, and shall be responsible for accomplishing the inspections and tests specified herein.

4.1.2 Previous Approval - Acceptance or approval of material during course of manufacture shall in no case be construed as a guaranty of the acceptance of the finished product.

4.1.3 Supervision of Tests - When inspection is conducted at the Contractor's plant, all inspection and testing shall be under the supervision of a Government Inspector.

4.1.4 Record of Tests - The Contractor shall maintain a record available to the Inspector, showing the results of all Inspection Tests.

4.1.5 Laboratory Facilities - Contractors not having laboratory facilities satisfactory to the Inspector shall engage the services of a commercial testing laboratory acceptable to the Inspector.

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4.1.6 Presubmission Testing - No item, part, or complete equipment shall be submitted to the Government Inspector by the Contractor until it has been previously tested and inspected by the Contractor and found to comply to the best of his knowledge and belief with all-applicable requirements. With the consent of the Contractor and at the discretion of the Procuring Agency, this prior test and inspection may be participated in or witnessed by the Government Inspector with the object of eliminating the necessity of repeating such test and inspection after the equipment has been formally submitted to the Government Inspector.

4.1.7 Rejection and Retests - The equipment which has been rejected may be reworked or have parts replaced to correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished the Inspector. Units rejected after retest shall not be resubmitted without the specific approval of the Procuring Agency.

4.2 Inspection

4.2.1 Examination of Product - Each assembly, subassembly, and part shall be examined as the Inspector may deem necessary to determine conformance with this specification and the applicable drawings with respect to material, dimensions, instructions, and workmanship.

4.3 Test Procedures

4.3.1 Classification - Test shall be classified as follows:

a. Type Tests
b. Inspection Tests

4.3.2 Type tests shall be performed on one (1) of the first twenty (20) production units and on one (1) of each subsequent group of one hundred (100) production units or parts thereof, for each manufacturer supplying equipment in accordance with this specification.

4.3.2.1 Type test samples shall be selected at random from production unit that have passed all inspection tests and shall have inspection tests repeated under environmental Conditions as specified in 3.8 of this specification.

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4.3.3 Inspection Tests - Inspection tests shall be made on all production assemblies submitted for acceptance under the Contract, to determine conformance with the requirements of this specification and to determine that the assemblies are. equivalent in performance and construction to the approved samples. All production lots as defined in 4.3.2 will be accepted contingent upon type tests approval.

4.3.3.1 Test Equipment - The following listed equipment will be required to perform tests specified in 4.3.3.2.

4.3.3.1.1 Control box wired in accordance with Figure 1.

4.3.3.1.2 Test stand. See Figure 2.

4.3.3.1.3. Spring scale.

4.3.3.1.4 Source of power, 115 volts, 400 cycles.

4.3.3.2 Tests - All units shall be subjected to tests specified in 4.3.3.2.1 through 4.3.3.2.1.5.2 at normal room conditions.

4.3.3.2.1 Torque and Switch .Contact Position Test

4.3.3.2.1.1 With control switches A, B, C, of Figure 1, in position 1, move the C-shaped lever, BuOrd drawing 760038, into its neutral position, and check for compliance with 3.7.1.1.

4.3.3.2.1.2 With control switches A, B, and c in position 1, move the C-shaped lever 0.046 ± .005 inch in both directions from its neutral position and.check for compliance with 3.7.1.2..

4.3.3.2.l.3 With control stitches A, B, and C in position 1, the indicator lamp, 505, of Figure 1, shall be illuminated.

4.3.3.2.1.4 With control switches A, B, and C in position 2, move the straight lever, BuOrd drawing 760040 into its neutral position, and check for compliance with 3.7.1.3.

4.3.3.2.1.5 With control switches A, B, and C in position 2, move the straight lever 0.125 ± .005 inch in both directions (towardS and away from the nameplate, BuOrd drawing 776416) from its neutral position, and check for compli ante with 3.7.1.4.

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4.3.3.2.1.5.1 With control switches A, B, and C in position 2, the indicator lamp, 506, Figure 1, shall light when the straight lever is moved 0.125 ± .005 inch towards the nameplate.

4.3.3.2.1.5.2 With the control switches A, B, and C in position 2, the indicator lamp, 506, shall not light when the straight lever. is moved 0.125 ± .005 inch in a direction away from the nameplate.

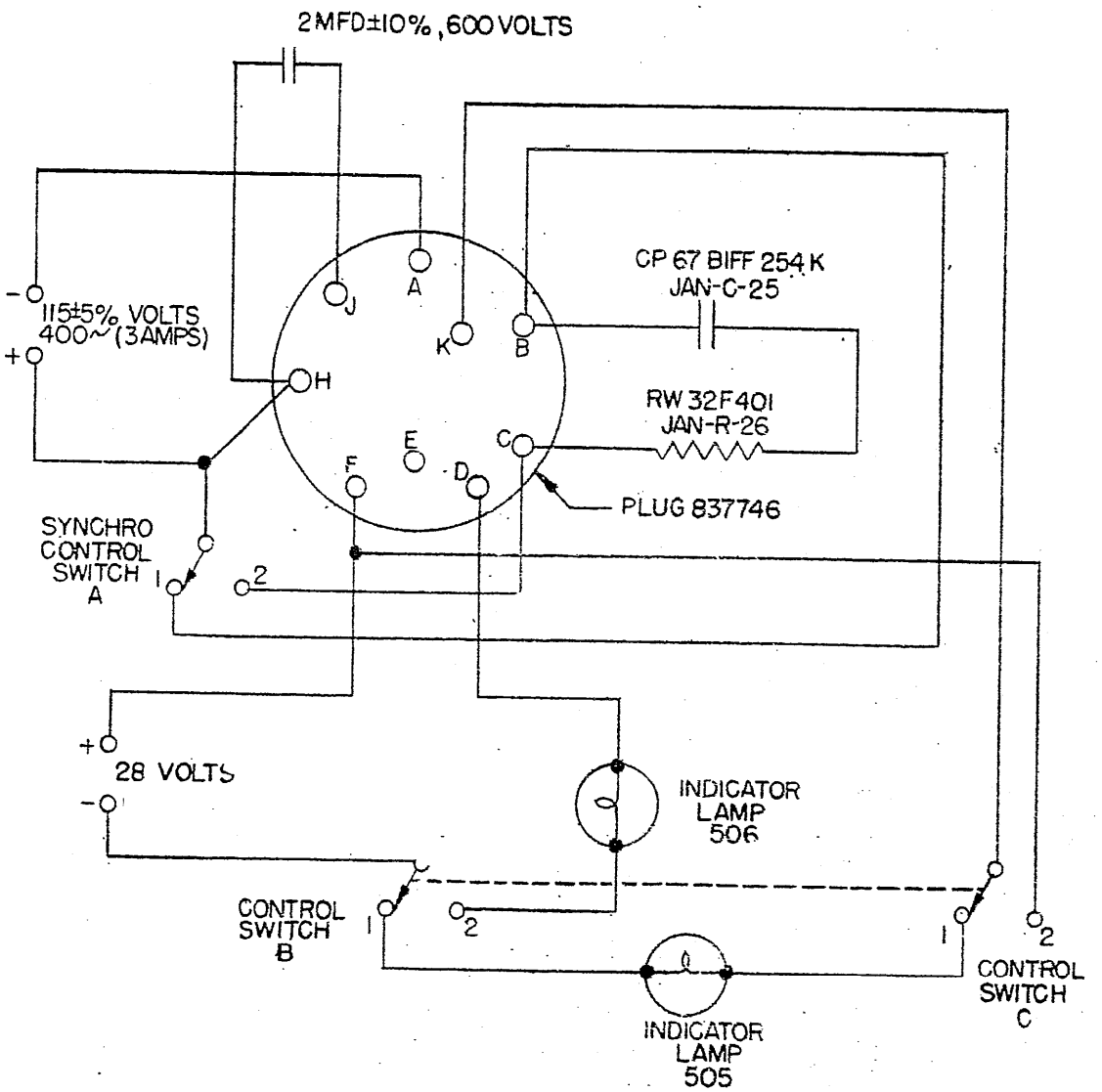


Figure 1. Test Control Circuit

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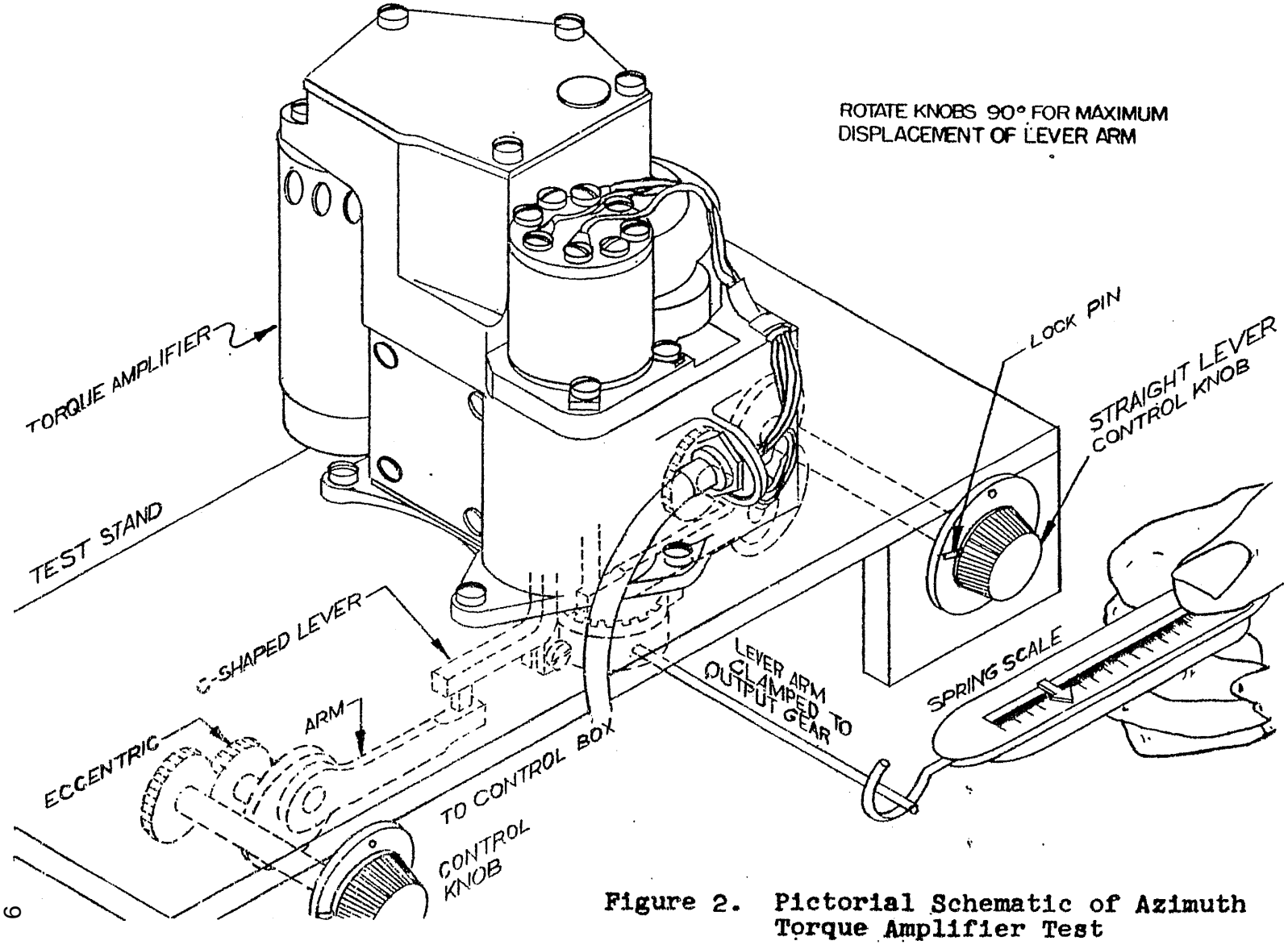


Figure 2. Pictorial Schematic of Azimuth Torque Amplifier Test

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5. PREPARATION FOR DELIVERY

5.1 Packaging and Packing - Packaging and packing shall be in accordance with MIL-P-116, Method II, paragraph 3.5.7.2.2 or 3.5.7.2.4, as applicable. Such items shall be packaged by either of the above methods as directed by the Procuring Agency. Further, metal containers shall be surrounded by suitable cushioning material and packed in a wooden case as directed by the Procuring Agency.

5.1.l Labeling and Marking - Labeling and marking shall be in accordance with MIL-STD-129, Military Standard Marking of Shipments, except as may be-modified by the Procuring Agency.

6. NOTES

6.1 Notice - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

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